

Physics assignment 1

1. The air bubble formed by explosion inside water performs oscillations with time period T which depends on pressure (p), density (ρ), and on energy due to explosion (E). By using dimension analysis method, establish relation between T , p , E and ρ .

2. $x(t) = t^2 + 2t - 4$ [meter]

Find: a) velocity at any time and velocity at ($t=2s$), [m/s]

b) acceleration at any time $a(t=2s)$, [m/s²]

3. $a(t) = t + 2$ [m/s²]

Find: a) Find velocity at any time

b) displacement at any time

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